



Sampling

Iqra Abdullah

Book Reference: Sekaran, Uma, and Roger Bougie. Research methods for business: A skill building approach. John Wiley & Sons, 2016.



Sampling

The process of selecting the right individuals, objects, or events as representatives for the entire population.

Example: In testing the life of bulbs, if we were to burn every single bulb produced, there would be none left to sell!

Why should we do sampling?

- Cost Effective
- Efficiency
- Feasibility
- Accuracy
- Better Understanding



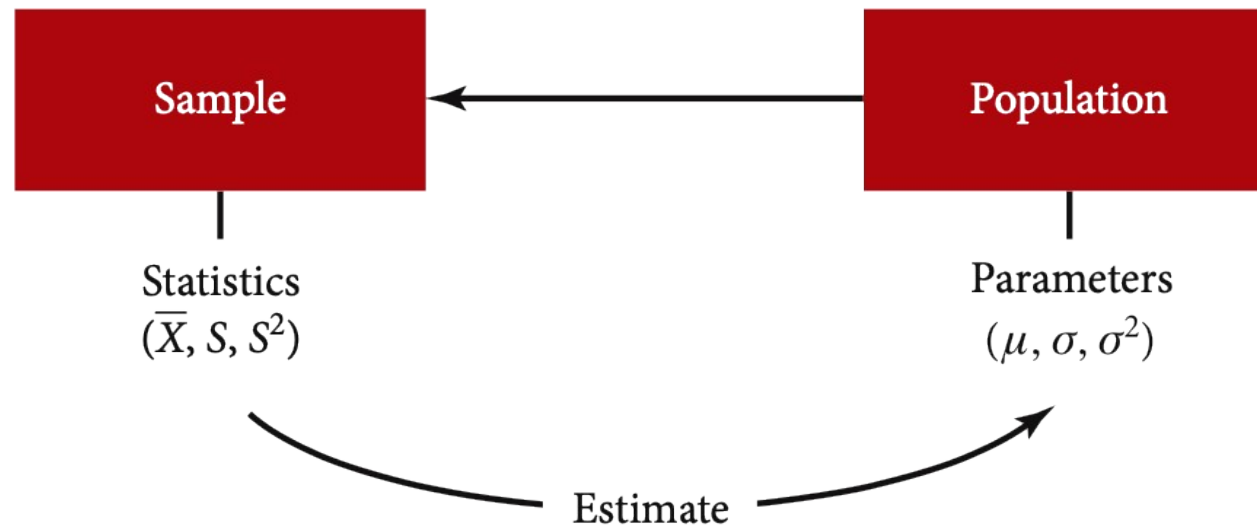


Related Terms

- ✓ **Population:** It is the group of people, events, or things of interest for which the researcher wants to make inferences (based on sample statistics).
 - ✓ **Element:** a single member of the population.
- **Sample:** a subset of the population
 - **Subject:** a single member of the sample

Statistical Terms for Sample and Population

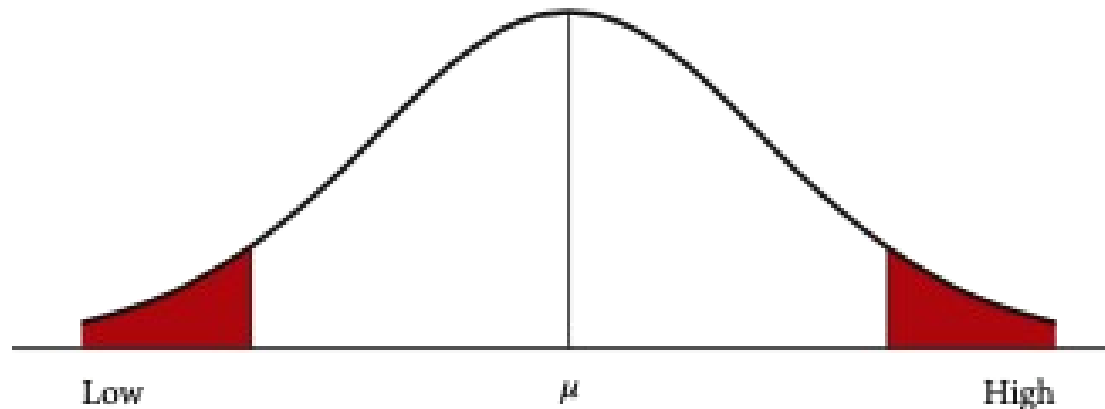
- **Parameters:** characteristics of the population such as μ (the population mean), σ (the population standard deviation), and σ^2 (the population variance)
- **Approximations:** The central tendencies, the dispersions, and other statistics in the sample of interest to the research e.g. $\bar{X}$ (the sample mean), S (the standard deviation), and S^2 (the variation in the sample)
- **Representativeness of Samples:** the sample statistic (e.g., $\bar{X}$, S , or S^2) is fairly close to the population parameter (i.e., μ , σ , or σ^2).

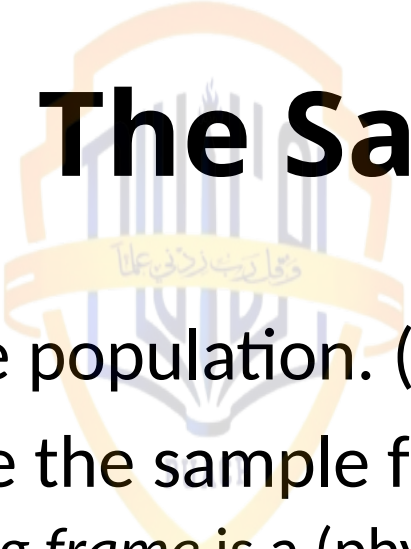


Normal Distribution

When attributes such as height and weight are considered, most people will be clustered around the mean, leaving only a small number at the extremes who are either very tall or very short, very heavy or very light, and so on.

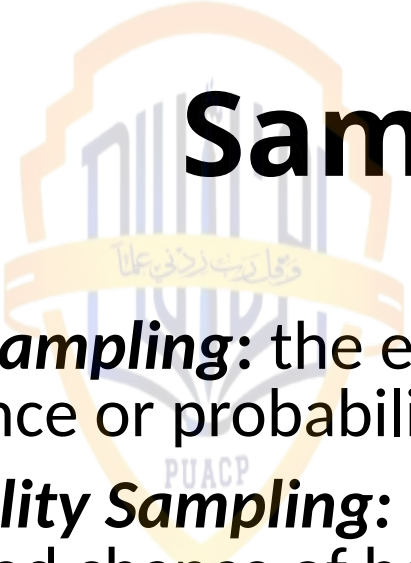
Sample size n increases, the means of the random samples taken from practically any population approach a normal distribution with mean μ and standard deviation σ .





The Sampling Process

1. Define the population. (elements, geographical boundaries etc.)
2. Determine the sample frame.
The *sampling frame* is a (physical) representation of all the elements in the population from which the sample is drawn.
3. Determine the sampling design.
4. Determine the appropriate sample size.
5. Execute the sampling process.



Sampling Design

- **Probability Sampling:** the elements in the population have some known, nonzero chance or probability of being selected as sample subjects
 - **Non-Probability Sampling:** the elements do not have a known or predetermined chance of being selected as subjects.
-
- What is the relevant target population of focus to the study?
 - What exactly are the parameters we are interested in investigating?
 - What kind of a sampling frame is available?
 - What costs are attached to the sampling design?
 - How much time is available to collect the data from the sample?



Sample Size

- The research objective.
- The extent of precision desired (the confidence interval).
- The acceptable risk in predicting that level of precision (confidence level). The amount of variability in the population itself.
- The cost and time constraints.
- In some cases, the size of the population itself.

Example 1: Population and Sample Size?

A satisfaction survey was conducted for a computer retailer in New Zealand. The objective of this survey was to improve internal operations and thus to retain more customers. Service satisfaction and several related variables were measured following a service encounter (i.e., a visit to the retailer). Hence, customer feedback was obtained while the service experience was still fresh. To obtain a representative sample of customers of the *computer retailer*, every tenth person, leaving one out of randomly selected stores in randomly selected cities, in randomly selected regions, was approached during a one-week period. Trained interviewers that were sent out with standardized questionnaires approached 732 customers leaving the stores.

Example 2: Population and Sample Size?

A young researcher was investigating the salesperson performance. To examine his hypotheses, data were collected from chief sales executives in the United Kingdom via mail questionnaires. The sample was drawn from a published business register. Before distributing the questionnaires, the young researcher called each selected company to obtain the name of the chief sales executive, who was contacted and asked to participate in the study. The questionnaires were subsequently distributed to chief sales executives of 450 companies. To enhance the response rate, pre-addressed and stamped envelopes were provided, anonymity was assured, and a summary of the research findings as an incentive to the participants was offered. Several follow-up procedures, such as telephone calls and new mailings, were planned in order to receive as many responses as possible.

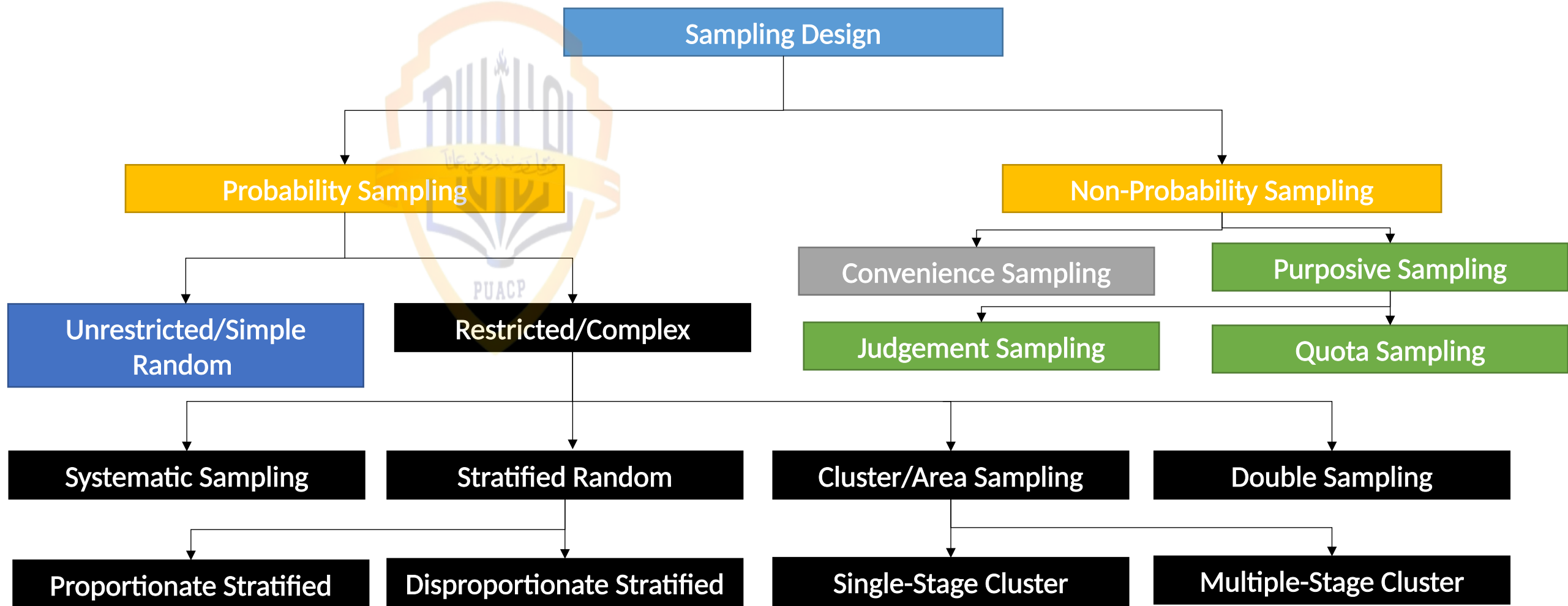


Exercise

I would like to assess the impact of a new teaching method on student performance in Akhuwat BS Students.

Determine following:

1. Define the population. (elements, geographical boundaries etc.)
2. Determine the sample frame (physical representation of all the elements in the population from which the sample is drawn)
3. Determine the sampling design. (Probability/Non-Probability)
4. Determine the appropriate sample size.
5. Execute the sampling process.





PROBABILITY SAMPLING

Unrestricted/Simple Random Sampling

- Every element in the population has a known and equal chance of being selected as a subject.

Example: Let's say an economist wants to study the average income of households in a particular city. Using a random selection method, the economist selects 100 households from the sampling frame.

Excel: RANDBETWEEN (__,__)

Systematic Random Sampling

- The systematic sampling design involves drawing every n th element in the population starting with a randomly chosen element between 1 and n .

Example:

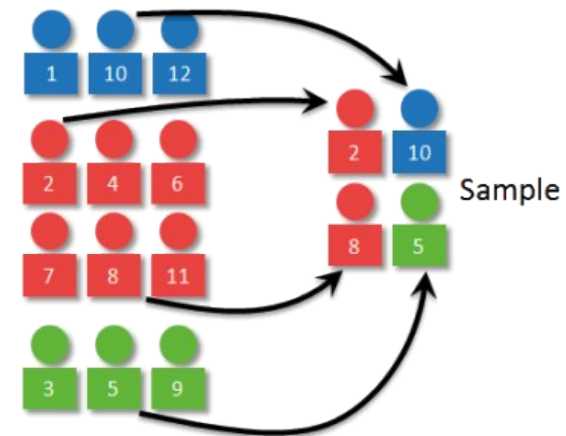
If we wanted a sample of 35 households from a total population of 260 houses in a particular locality, then we could sample every seventh house starting from a random number from 1 to 7. Let us say that the random number was 7, then houses numbered 7, 14, 21, 28, and so on, would be sampled until the 35 houses were selected.

Stratified Random Sampling

- Stratified random sampling, as its name implies, involves a process of stratification or segregation, followed by random selection of subjects from each stratum.

Examples:

1. Are the machine operators more accident prone than clerical workers?
2. Are Pakistani workers more loyal to the organization than Americans?



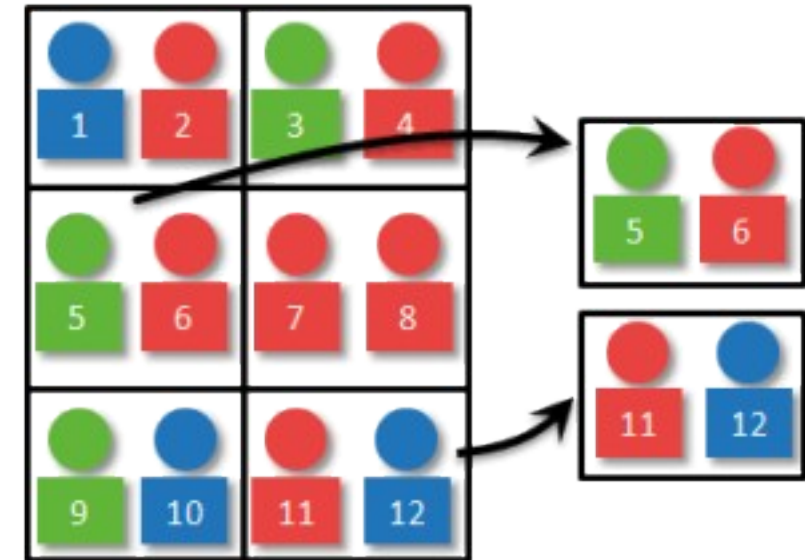
Proportionate and Disproportionate Stratified Random Sampling

Job level	Number of elements	Number of subjects in the sample	
		Proportionate sampling (20% of the elements)	Disproportionate sampling
Top management	10	2	7
Middle-level management	30	6	15
Lower-level management	50	10	20
Supervisors	100	20	30
Clerks	500	100	60
Secretaries	20	4	10
Total	710	142	142

Cluster Sampling

- In cluster sampling, the target population is first divided into clusters. Then, a random sample of clusters is drawn and for each selected cluster either all the elements or a sample of elements are included in the sample. Cluster samples offer more heterogeneity within groups and more homogeneity among groups.
- Cluster sampling can also be done in several stages and is then known as multistage cluster sampling or in one stage known as one-stage sampling.

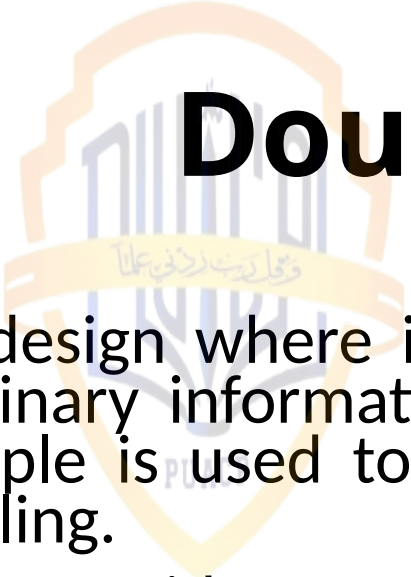
Example: A telephone company wants to install a public telephone outlet in a locality where crime is most rampant, so that victims can have access to a telephone. Studying the crime statistics and interviewing the residents in a particular area will help to choose the right location for installation of the phone.



The logo of the Punjab University of Agriculture, Chaudhary (PUACP) is positioned behind the title. It features a shield with a book and the acronym PUACP.

Difference Between Stratified/Cluster Sampling

<https://www.youtube.com/watch?v=O5TjsbZBs2E>



Double Sampling

- A sampling design where initially a sample is used in a study to collect some preliminary information of interest, and later a subsample of this primary sample is used to examine the matter in more detail, is called double sampling.

Example: Let's consider a government agency that wants to estimate the unemployment rate in a large country. Due to budget constraints and the vastness of the population, it's not feasible to conduct a comprehensive survey of all households at once. Instead, they decide to use double sampling:

- First Stage - The country is divided into regions or clusters, such as states or counties. A random sample of these regions is selected.
- Second Stage - Within each selected region, a further random sample of households is selected.



NON-PROBABILITY SAMPLING

Convenience Sampling

- Convenience sampling refers to the collection of information from members of the population who are conveniently available to provide it.

Example: A contest, with the purpose of determining whether people prefer one product to another, might be held at a shopping mall visited by many shoppers.



Purposive Sampling

The sampling here is confined to specific types of people who can provide the desired information, either because they are the only ones who have it, or they conform to some criteria set by the researcher.

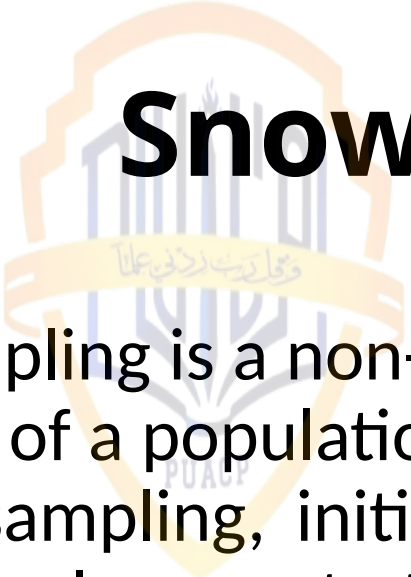
- **Judgement Sampling:** the sampling design is used when a limited number or category of people have the information that is sought.

Example: A researcher wants to find out what it takes for women managers to make it to the top, the only people who can give first-hand information are the women who have risen to the positions of presidents, vice presidents, and important top-level executives in work organizations.

Purposive Sampling

- **Quota Sampling:** Quota sampling, a second type of purposive sampling, ensures that certain groups are adequately represented in the study through the assignment of a quota.

Example: it may be surmised that the work attitude of blue-collar workers in an organization is quite different from that of white-collar workers. If there are 60% blue-collar workers and 40% white-collar workers in this organization, and if a total of 30 people are to be interviewed to find the answer to the research question, then a quota of 18 blue-collar workers and 12 white-collar workers will form the sample, because these numbers represent 60% and 40% of the sample size. The first 18 conveniently available blue-collar workers and 12 white-collar workers will be sampled according to this quota.



Snowball Sampling

Snowball sampling is a non-probabilistic sampling technique used when the members of a population of interest are difficult to locate or access. In snowball sampling, initial participants are selected based on their availability or relevance to the research topic, and then they refer other potential participants. This process continues like a snowball rolling down a hill, with each participant referring additional participants, hence the name "snowball sampling."



Sampling design	Description	Advantages	Disadvantages
Probability sampling			
1. Simple random sampling	All elements in the population are considered and each element has an equal chance of being chosen as the subject.	High generalizability of findings.	Not as efficient as stratified sampling.
2. Systematic sampling	Every n th element in the population is chosen starting from a random point in the sampling frame.	Easy to use if sampling frame is available.	Systematic biases are possible.

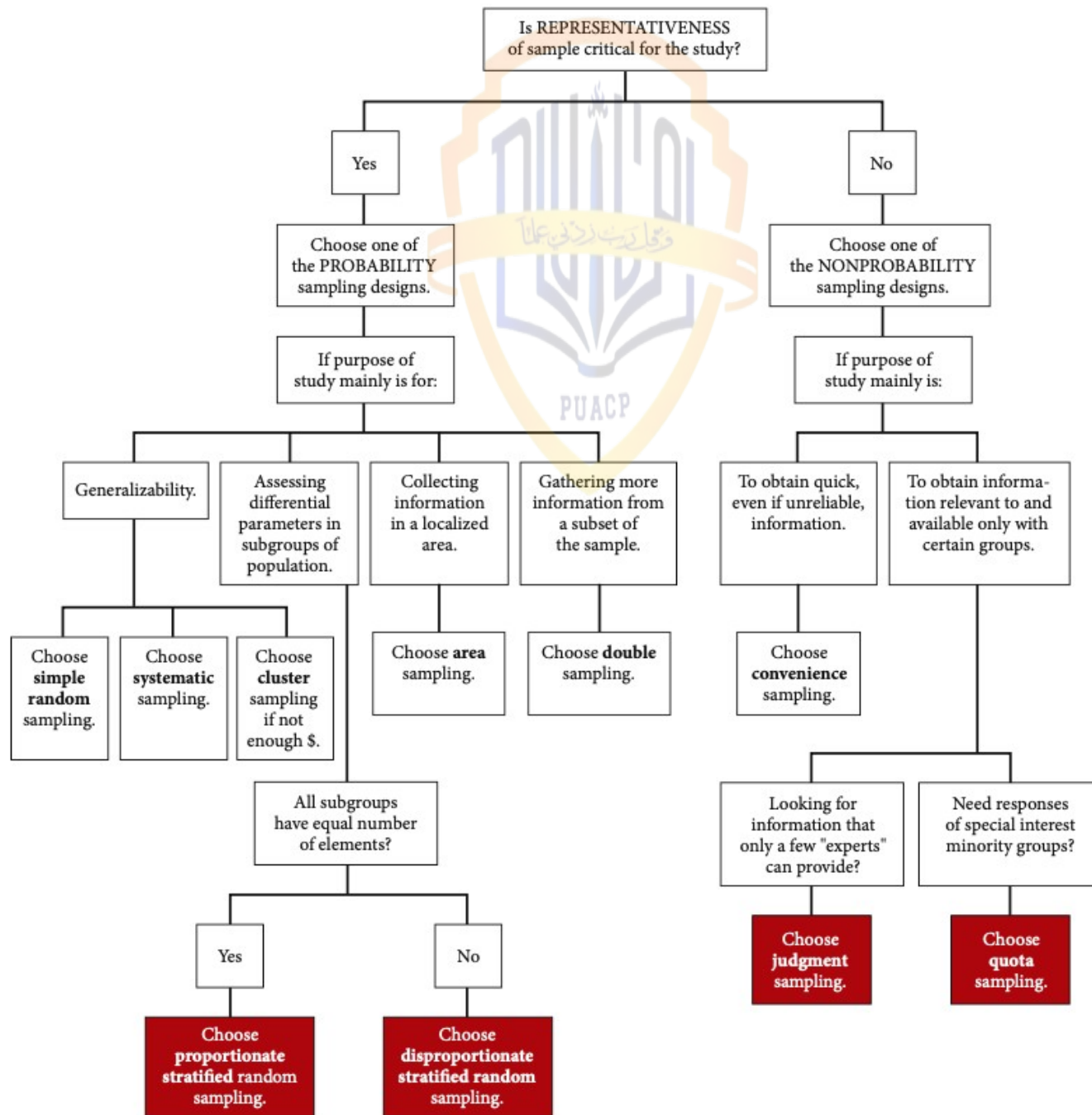
Sampling design	Description	Advantages	Disadvantages
3. Stratified random sampling (Str.R.S.) Proportionate Str.R.S. Disproportionate Str.R.S.	Population is first divided into meaningful segments; thereafter subjects are drawn in proportion to their original numbers in the population. Based on criteria other than their original population numbers.	Most efficient among all probability designs. All groups are adequately sampled and comparisons among groups are possible.	Stratification must be meaningful. More time consuming than simple random sampling or systematic sampling. Sampling frame for each stratum is essential.
4. Cluster sampling	Groups that have heterogeneous members are first identified; then some are chosen at random; all the members in each of the randomly chosen groups are studied.	In geographic clusters, costs of data collection are low.	The least reliable and efficient among all probability sampling designs since subsets of clusters are more homogeneous than heterogeneous.



Sampling design	Description	Advantages	Disadvantages
5. Area sampling	Cluster sampling within a particular area or locality.	Cost-effective. Useful for decisions relating to a particular location.	Takes time to collect data from an area.
6. Double sampling	The same sample or a subset of the sample is studied twice.	Offers more detailed information on the topic of study.	Original biases, if any, will be carried over. Individuals may not be happy responding a second time.



Sampling design	Description	Advantages	Disadvantages
Nonprobability sampling			
7. Convenience sampling	The most easily accessible members are chosen as subjects.	Quick, convenient, less expensive.	Not generalizable at all.
8. Judgment sampling	Subjects selected on the basis of their expertise in the subject investigated.	Sometimes, the only meaningful way to investigate.	Generalizability is questionable; not generalizable to entire population.
9. Quota sampling	Subjects are conveniently chosen from targeted groups according to some predetermined number or quota.	Very useful where minority participation in a study is critical.	Not easily generalizable.

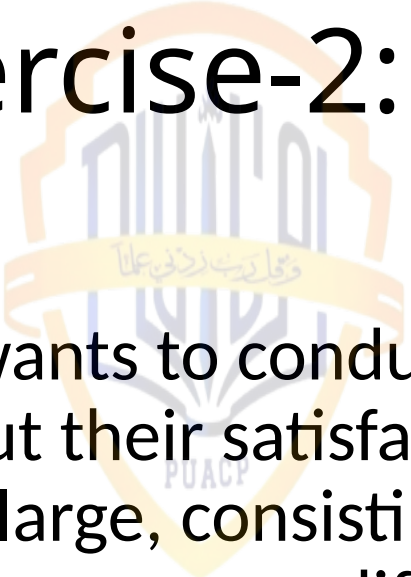


How to choose sampling design?

Exercise-1: Simple Random Sampling

A city council is conducting a survey to assess public opinion on a proposed new public transportation system. They want to ensure that the survey accurately represents the views of all residents across the city. They have a comprehensive list of all residential addresses within the city limits.

Question: Which sampling method would you recommend, and why?



Exercise-2: Systematic Random Sampling

A university wants to conduct a survey to gather feedback from students about their satisfaction with campus facilities. The student population is large, consisting of thousands of students enrolled in various programs across different departments. The university administration wants to ensure that the survey captures the opinions of students from all departments and academic levels.

Question: Which sampling method would you recommend, and why?



Exercise-3

A public health department wants to assess the prevalence of a particular health condition in Lahore city. The city is divided into several neighbourhoods, each with its own distinct characteristics and demographics. The health department wants to ensure that their study captures a representative sample of the city's population while minimizing costs and logistical challenges.

Question: Which sampling method would you recommend, and why?



Exercise-4

A city's Department of Education wants to assess the academic performance of students in public schools across different grade levels (elementary, middle, and high school). The city has a total of 30 public schools, with varying student populations in each school. The department wants to ensure that the sample represents each grade level proportionately to their overall enrollment in the city.

Question: Which sampling method would you recommend, and why?



Exercise-5

A marketing research firm is conducting a study to assess the preferences of consumers for a new line of smartphones in a large metropolitan area. They want to ensure that the sample includes a diverse range of demographic groups, including age, income level, and education level. However, they are particularly interested in oversampling certain demographic groups that they believe are key target markets for the new smartphones, such as young professionals with high disposable income.

Question: Which sampling method would you recommend, and why?



Exercise-6

A high school newspaper club wants to gather opinions from students about their favourite extracurricular activities for an upcoming article. Due to time constraints and limited resources, they decide to approach students who are readily available in the school cafeteria during lunchtime.

Question: Which sampling method would you recommend, and why?



Exercise 7

A market research firm is conducting a study to assess the preferences of consumers for a new brand of energy drinks. They want to ensure that their sample represents a diverse range of demographic groups, including age, gender, income level, and occupation. However, they have limited resources and time constraints, making it impractical to use random sampling methods.

Question: Which sampling method would you recommend, and why?



Exercise 8

You are a researcher working for a market research firm tasked with assessing customer satisfaction with a new online shopping platform. Due to budget constraints and time limitations, you have decided to use double sampling to collect data from the target population efficiently.

Task: How would you conduct the double sampling.